

Title

Efficacy and safety of alofropodect in acute schizophrenia: a phase 2 randomized controlled trial

Abstract

Background: Phosphodiesterase 10A inhibition represents a novel mechanistic approach to psychotic symptom management, though prior agents in this class have shown limited clinical translation. Alofropodect exhibits improved pharmacological characteristics and brain penetration. **Objective:** To assess efficacy and safety of alofropodect for acute schizophrenia in hospitalized patients. **Methods:** A total of 189 patients (mean age 41.8 years; 63% male) with acute schizophrenia were randomly assigned 1:1:1 to alofropodect 20 mg daily, alofropodect 40 mg daily, or placebo for 4 weeks. Primary endpoint was PANSS positive subscale score change at week 4. Secondary endpoints included PANSS total and negative subscales, clinical response ($\geq 30\%$ PANSS reduction), and adverse event rates. **Results:** The 40 mg dose showed primary endpoint success (difference from placebo -6.4 points, $P = 0.001$); the 20 mg dose demonstrated borderline significance (-3.8 points, $P = 0.048$). Clinical response rates were 41% (40 mg), 15% (20 mg), and 4% (placebo). Both doses improved PANSS negative subscale scores ($P < 0.05$). Extrapyramidal symptoms occurred in 8% of treatment recipients versus 0% placebo. Somnolence was reported by 11% receiving active drug. Metabolic parameters remained stable. No treatment-related serious adverse events were observed. **Conclusion:** Alofropodect, particularly at the 40 mg dose, demonstrates efficacy in acute schizophrenia with a tolerable side-effect profile supporting phase 3 evaluation.